Google 🔍 Ayush Singh - distributed systems engineer ✕ ⚙ ⋮⋮⋮ You

All | Experience | Projects | ✦ AI Mode | Skills | More | Tools

ayushsingh.is-a.dev › work-experience › distributed-systems-engineer

✦ AI Overview · Executive Summary

Software Engineer III (L4) at **Google** (Munich) with 4+ years building planet-scale **distributed systems**, **low-latency** C++/Java RPC frameworks, fault-tolerant distributed file systems, and real-time streaming pipelines across Google, Komprise, Accenture, and Tookitaki.

- **Planet-Scale Core Infra:** Defaulted architecture across ~5,200 microservice nodes (Search, YouTube, Maps) & built distributed storage serving 100,000+ services.
- **High-Impact Systems ROI**: Delivered $2.5M+/quarter savings (SLA platform) and 94% streaming latency reduction (850ms → <50ms).
- **Fast Track Record**: Promoted L3 → L4 in 15 months; VP Spot Bonus ; 3× Readability (C++, TS, API) & 1040+ PRs reviewed.

Top stats

~5,200	100K+	$2.5M+	94%
Prod microservice nodes running my default design	Google services backed by my distributed FileSystem	Quarterly savings via SLA/SLO platform	Streaming latency cut (850ms → <50ms)

Work Experience · 4+ Years in Distributed Systems, Storage & Backend

Google — Software Engineer III (L4), Distributed Systems & Core Infrastructure

Munich, Germany · Nov 2024 – Present · C++ · Java · Kotlin · gRPC · Distributed Microservices · Edge Routing

- **Core C++ & Java Request Frameworks:** Shipped global request-path processors into **Google's core Java backend framework and C++ Search engine, sustain**ing 7 billion+ daily requests across Search, YouTube, and Maps.
- **SignedOut Architecture Across 5,200+ Nodes**: Standardized the Google wide SignedOut request-processing architecture defaulted across **~5,200 production microservice nodes** (210+ modules), cutting signal-processing complexity by 8**%** and retiring **1,300+ legacy LOC**.
- **Cross-Product Telemetry & Batch Compute:** Built parallel stream shadow-diffing across **YouTube, Play, Search & Payments** (100% live migration parity across 10B+ daily RPCs) and migrated 7 critical identity pipelines to batch compute (12% lower p99 latency, reclaiming GCUs).
- **High-Velocity Regulatory Rollouts:** Led end-to-end **UK Online Safety Act & US AppStore** distributed backend launches across Web, Android & iOS in 4 weeks across 5+ teams (VP Spot Bonus).

Google — Software Engineer II → III (L4), Distributed Storage & Observability

Hyderabad, India · Jul 2022 – Nov 2024 · C++ · Go · FUSE · Distributed File Systems · Paxos Consensus · OpenTelemetry

- **Distributed File System (100K+ Services):** Engineered fault-tolerant storage and replication for **Google's internal HDFS-like distributed file system**, optimizing mission-critical binary delivery across **100,000+ internal services**.
- **Overload Protection & Consensus Locks:** Shipped server-side admission control & load-shedding, semaphore RPC concurrency limiting in the **FUSE client**, and migrated distributed snapshot locks off Paxos lock services, cutting lock contention by 65%.
- **$2.5M+ Quarterly Savings via SLA/SLO Platform:** Architected a highly available observability & automated **SLA/SLO compliance platform** correlating distributed traces and metrics, saving **$2.5M+ per quarter**.
- **OpenTelemetry Collector & Go Gateway:** Co-architected a trace collector over **gRPC-Web** & encrypted Go asset gateway (50K+ req/s); led Chrome 3P cookie deprecation across 120+ services (CEO approved) & promoted to L4 in 15 months.



Ayush Singh

Distributed Systems Engineer · Google L4

Overview | Experience | Skills

Current: SWE III (L4) at Google
Experience: 4+ Years
Location: Munich, Germany
Email: apparently.ayush@gmail.com
Phone: +49 1735 34 8915
Education: NIT Agartala · B.Tech EE (CGPA: 9.32 / 10)
Code Authority: 3× Google Readability (C++, TS, API) · 1,040+ Reviews
Honors: VP Spot Bonus · 7× Peer Bonuses · GFG Rank #72

Profiles & Links

LinkedIn | Portfolio | GitHub | Email

Technical skills

Languages
C++ Java Go TypeScript Python Scala Kotlin SQL

Distributed Systems & Storage
gRPC Protobuf Distributed FS (HDFS / FUSE) Spanner Paxos / Raft Consensus Apache Spark Kafka ActiveMQ Snowflake Redis

Cloud, Observability & AI Infra
Kubernetes Docker OpenTelemetry GCP AWS Terraform CI/CD WebGPU / LLM Agents

Komprise — Software Engineer Intern (Distributed Storage Infra)

Bengaluru, India · Jan 2022 – Jun 2022 · Python · Multi-Threading · Distributed Systems Testing · CI/CD

- Parallel Test Engine: Multi-threaded cluster verification cutting runtime by 87.5% & increasing bug coverage by 60%.

Accenture — Data Engineer Intern (Cloud Data Infra)

India · May 2021 – Jul 2021 · AWS S3 · Snowflake · Distributed ETL · Python · SQL

- 10TB+ Cloud ETL: Distributed AWS S3 to Snowflake pipelines cutting manual reporting time by 70%.

Tookitaki — Software Engineer Intern (Real-Time Distributed Streaming)

Singapore · Oct 2020 – Jan 2021 · Apache Spark Structured-Streaming · Scala · ActiveMQ · JMS

- 94% Latency Cut (850ms → 45ms): Real-time Apache Spark Structured-Streaming + ActiveMQ pipeline with connection pooling.

✦ Selected Open-Source & Agentic Systems

1. YTM AI Host — Real-Time (<100ms) AI Radio Jockey & Audio Streaming Engine · GitHub

TypeScript · Python · Gemini API · WebLLM / WebGPU · On-Device Inference · Real-Time Audio Pre-Fetching

- **Sub-100ms Streaming AI: Built a local** streaming server & Chrome extension for YouTube Music & Spotify combining on-device WebGPU/WebLLM inference and neural TTS for <100ms audio transitions.

2. LinkinJMS (High-Performance Spark Queue Connector) & Multi-Agent Incident Triage

- **LinkinJMS (Scala / Apache Spark / ActiveMQ):** Open-source bidirectional streaming connector achieving **<50ms message latency** with **exactly-once delivery**, AMQP/TLS encryption, and automatic failover.
- **Autonomous Multi-Agent Oncall Triage (Google):** Autonomous Python multi-agent system fanning out parallel queries across rollout logs, code changes, and metrics to email root-cause reports before on-call triage.

People also ask

What scale of distributed systems does Ayush build at Google?
His C++/Java middleware runs on the critical request path of Search, YouTube & Maps (~5,200 microservice nodes), and his distributed storage systems serve 100K+ services. ˅

How fast was Ayush promoted at Google?
Promoted L3 → L4 (SWE III) in 15 months with a VP of Engineering Spot Bonus, 7× peer bonuses, and 3× Google Readability certifications (C++, TypeScript, API; 1,040+ code reviews). ˅

What measurable systems impact has Ayush delivered?
$2.5M+/quarter saved via automated SLA/SLO enforcement, 94% streaming latency reduction (850ms → <50ms), 8% lower RPC complexity, and 87.5% faster distributed test execution. ˅

Where did Ayush study and what are his algorithmic credentials?
NIT Agartala (B.Tech EE, 9.32/10 CGPA). Ranked #1 in Institute & #72 Worldwide on GeeksForGeeks (top 0.07% of 100K+ developers), IASc Fellow, and Founding Google DSC Lead. ˅

People also search for

Distributed File Systems (HDFS/FUSE) · High-Throughput gRPC · Apache Spark Streaming · OpenTelemetry · Paxos & Consensus

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